

FOOD NUTRITION & HEALTH RISK ANALYSIS

Project Overview

The Food Nutrition & Health Risk Analytics System is a data analytics project designed to analyze the nutritional composition of various food items and identify potential health risks associated with their consumption.

The project utilizes Python for data cleaning and feature engineering, SQL for analytical querying, and Power BI for interactive dashboard development. The objective is to transform raw nutrition data into meaningful insights that can support healthier food choices and nutrition-based decision-making.

Objectives

The main objectives of this project are:

- Analyze nutritional characteristics of food items.
 - Identify foods with high potential health risks.
 - Evaluate nutritional quality using a Nutrition Density metric.
 - Perform analytical queries using SQL.
 - Build an interactive Power BI dashboard for data visualization.
 - Generate actionable insights from food nutrition data.
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Dataset Description

The dataset contains nutritional information for a variety of food items.

Key Attributes Used

Column	Description
Food Item	Name of the food
Energy (kcal)	Total calories
Carbs	Carbohydrate content
Protein (g)	Protein content

Column	Description
Fat (g)	Fat content
Free Sugar (g)	Sugar content
Fibre (g)	Dietary fibre
Cholesterol (mg)	Cholesterol content
Calcium (mg)	Calcium content

The dataset was processed and transformed to derive additional metrics for health analysis.

Data Cleaning & Feature Engineering

Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Handled missing values.
- Standardized column names.
- Verified data types for numerical analysis.
- Checked data consistency across nutritional attributes.

Feature Engineering

Health Risk Score

A custom Health Risk Score was developed to estimate the potential health impact of food items.

The score considers nutritional factors such as:

- Fat Content
- Free Sugar Content
- Cholesterol Content

Higher scores indicate greater potential health risks.

Nutrition Density

Nutrition Density was created to evaluate the nutritional value of food relative to calorie content.

Higher Nutrition Density values indicate foods that provide more nutrients per calorie consumed.

Risk Classification

Food items were categorized into:

- Low Risk
- Moderate Risk
- High Risk

based on their calculated Health Risk Score.

SQL Analysis & Key Findings

To extract meaningful insights from the nutrition dataset, several SQL queries were performed on the processed data stored in MySQL. The analysis focused on identifying healthy foods, high-risk foods, nutrient-rich foods, and overall dietary patterns.

1. Which Foods Are the Most Nutrient-Dense?

Objective

Identify foods that provide the highest nutritional value relative to calorie content.

SQL Query:

```
SELECT food_item,  
nutrition_density  
FROM nutrition  
WHERE is_spice_condiment = FALSE  
ORDER BY nutrition_density DESC  
LIMIT 10;
```

SQL Query Result:

	food_item	nutrition_density
►	Maa chaane ki dal	58.93368100382349
	Cracked wheat and green gram...	50.41961608017326
	Sesame laddoo (Til ke laddoo)	49.718859794374396
	Dessicated coconut chutney (So...	48.435173315933156
	Lobster Roll Sandwich	39.188079457470465
	Pearl millet infant food (Bajra s...	38.24943740047132
	Besan bathua chilla/cheela	38.19380801786269
	Dhokla	37.892349477583
	Rice flakes and roasted channa	37.818723068623626
	Channa murmura premix	36.8877622623889

Figure 3: Top 10 Foods Ranked by Nutrition Density

Key Findings

- Maa Chaane Ki Dal achieved the highest nutrition density score in the dataset.
- Cracked Wheat and Green Gram Dal Premix ranked second.
- Traditional foods such as Dhokla and Soya Roti demonstrated excellent nutritional value.
- Foods rich in protein, fibre, and minerals generally achieved higher nutrition density values.

2. Which Foods Have the Highest Health Risk Scores?

Objective

Identify foods that may contribute to adverse health outcomes due to high levels of sugar, fat, or cholesterol.

SQL Query:

```
SELECT food_item,
health_risk_score
FROM nutrition
ORDER BY health_risk_score DESC
LIMIT 10;
```

SQL Query Result:

	food_item	health_risk_score
►	Butter icing	100
	Glace icing	95.71269147865063
	Royal icing	95.42513950402095
	Chocolate butter icing	86.49798367084635
	Mango murabba candy	84.84145937751695
	Chocolate glace icing	83.78901940199195
	American frosting	77.6057415759551
	Christmas star biscuit	71.50978333186849
	Dry mango chutney (Sookhe aa...	71.23104054482718
	Peanut burfi (Moongfali ki burfi)	69.3577354124589

Figure 4: Top 10 Foods with Highest Health Risk Scores

Key Findings

- Butter Icing recorded the highest health risk score.
 - Glace Icing and Royal Icing followed closely behind.
 - Several frosting and icing products appeared among the highest-risk foods.
 - Foods with high sugar content consistently generated elevated risk scores.
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3. Which Foods Are Rich in Fibre but Low in Free Sugar?

Objective

Identify foods that support digestive health while maintaining low sugar levels.

SQL Query:

```
SELECT food_item,  
       fibre_g,  
       free_sugar_g  
FROM nutrition  
WHERE free_sugar_g < (SELECT AVG(free_sugar_g) FROM nutrition)  
AND is_spice_condiment = FALSE  
ORDER BY fibre_g DESC  
LIMIT 10;
```

SQL Query Result:

	food_item	fibre_g	free_sugar_g
▶	Maa chaane ki dal	13.45	1.05
	Cracked wheat and green gram...	11.66	0.63
	Mustard seeds baghar (Mustard...	9.74	1.14
	Rice flakes and roasted channa	8.84	1.31
	Soya roti	8.71	1.58
	Lotus stem pickle (Kamal kakdi ...	8.58	2.53
	Namkeen daliya	8.57	0.82
	Channa murmura premix	8.21	1.52
	Black channa curry/Bengal gra...	7.99	2.36
	Besan bathua chilla/cheela	7.79	1.45

Figure 5: High-Fibre and Low-Sugar Foods

Key Findings

- Maa Chaane Ki Dal ranked first with the highest fibre content.
- Cracked Wheat and Green Gram Dal Premix also demonstrated excellent fibre levels.
- Soya Roti and Black Channa Curry emerged as strong choices for digestive health.
- Most top-performing foods were based on pulses, legumes, or whole grains.

4. Which Foods Have High Protein Content While Maintaining a Low Health Risk Score?

Objective

Identify healthier protein sources suitable for fitness and balanced nutrition.

SQL Query:

```
SELECT food_item,  
protein_g,  
health_risk_score  
FROM nutrition  
WHERE health_risk_score < 40  
AND is_spice_condiment = FALSE  
ORDER BY protein_g DESC  
LIMIT 10;
```

SQL Query Result:

	food_item	protein_g	health_risk_score
►	Roast chicken	20.49	16.243793632020704
	Lemon chicken	20.2	13.958010015769792
	Lobster Roll Sandwich	20	28.47092577797992
	Maa chaane ki dal	19.8	9.270049947089166
	Lemon butter fish	16.38	14.568420991902665
	Tandoori chicken	16.26	14.515346616425104
	Baked stuffed fish	16.14	11.674730441704462
	Fish tikka	16	11.271334526724228
	Cracked wheat and green gram...	15.89	9.291273249752088
	Afghani chicken	15.66	14.573931123175132

Figure 6: High-Protein Foods with Low Health Risk

Key Findings

- Roast Chicken achieved the highest protein content among low-risk foods.
- Lemon Chicken, Tandoori Chicken, and Fish Tikka ranked among the best animal-based protein sources.
- Maa Chaane Ki Dal and Cracked Wheat with Green Gram Premix emerged as strong plant-based alternatives.
- Both animal and plant-based foods demonstrated the ability to provide high-quality protein while maintaining relatively low health risk scores.

5. What Percentage of Foods Fall into Low, Medium, and High Health-Risk Categories?

Objective

Analyze the overall health profile of foods within the dataset.

SQL Query:

```
SELECT CASE  
WHEN health_risk_score <= 33 THEN 'Low Risk'  
WHEN health_risk_score <= 66 THEN 'Medium Risk'  
ELSE 'High Risk'  
END AS risk_category,  
COUNT(*)  
FROM nutrition  
WHERE is_spice_condiment = FALSE  
GROUP BY risk_category;
```

SQL Query Result:

	risk_category	COUNT(*)
►	High Risk	12
	Low risk	662
	Medium risk	346

Figure 7: Health Risk Category Distribution

Key Findings

- The majority of foods belong to the Low-Risk category.
- Only a small percentage of foods were classified as High Risk.
- Most High-Risk foods consisted of desserts, confectionery products, and sugar-rich preparations.

6. What Are the Top 10 Healthiest Foods Overall?

Objective

Rank foods by combining Nutrition Density and Health Risk Score into a single overall health metric.

SQL Query:

```
SELECT food_item,  
nutrition_density,  
health_risk_score,  
(nutrition_density - health_risk_score) AS overall_health_score  
FROM nutrition  
WHERE is_spice_condiment = FALSE  
ORDER BY overall_health_score DESC  
LIMIT 10;
```

SQL Query Result:

	food_item	nutrition_density	health_risk_score	overall_health_score
▶	Maa chaane ki dal	58.93368100382349	9.270049947089166	49.66363105673432
	Cracked wheat and green gram...	50.41961608017326	9.291273249752088	41.128342830421175
	Dessicated coconut chutney (So...	48.435173315933156	21.94360014302585	26.491573172907305
	Rice flakes and roasted channa	37.818723068623626	12.078808688469234	25.739914380154392
	Channa murmura premix	36.8877622623889	12.824473314845315	24.063288947543583
	Soyabean curry	36.37924690651742	12.789976870814474	23.58927003570295
	Dhokla	37.892349477583	14.540036507996826	23.352312969586173
	Soya roti	36.716669906434156	13.998159965431281	22.718509941002875
	Consomme au vermicelli	33.06301727358129	10.461823903076144	22.601193370505143
	Meat consomme (with mutton)	32.96415913229073	10.38031709264213	22.5838420396486

Figure 8: Top 10 Healthiest Foods Overall

Key Findings

- Maa Chaane Ki Dal ranked as the healthiest food in the dataset.
 - Cracked Wheat and Green Gram Dal Premix secured the second position.
 - Dhokla, Soya Roti, and Soyabean Curry consistently appeared among the healthiest options.
 - Foods with high nutrition density and low health risk scores dominated the rankings.
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Overall SQL Insights

The SQL analysis revealed several important trends:

- Foods with high nutrition density generally provide superior nutritional value per calorie consumed.
- High health risk scores are strongly associated with foods rich in sugar and unhealthy fats.
- Pulse-based, grain-based, and legume-based foods consistently appeared among the healthiest options.
- Several protein-rich foods maintained low health risk scores, making them suitable for balanced dietary planning.
- Combining Nutrition Density and Health Risk Score proved effective in identifying foods that maximize nutritional benefits while minimizing potential health concerns.

These findings formed the foundation for the Power BI dashboard and support data-driven food health assessment and dietary decision-making.

Power BI Dashboard

An interactive Power BI dashboard was developed to visualize nutritional trends and health indicators.

Dashboard Features

- KPI Cards
 - Total Foods
 - Average Health Risk Score
 - Average Nutrition Density
 - Percentage of Low-Risk Foods
- Risk Category Distribution

- Top Healthy Foods
- Top High-Risk Foods
- Nutrition Analysis
- Interactive Filtering using Slicers

Dashboard Benefits

The dashboard enables users to:

- Quickly identify healthy food options.
- Compare nutritional profiles.
- Explore health risks associated with food consumption.
- Make data-driven dietary decisions.



Key Insights

The analysis revealed several important findings:

- Foods with high free sugar and fat content contribute significantly to health risk scores.
 - Nutrient-dense foods often contain higher levels of protein and fibre.
 - Many foods with moderate calorie values differ substantially in nutritional quality.
 - Risk categorization provides a simple method for evaluating food healthiness.
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Business Impact

This project demonstrates how data analytics can be applied in the health and nutrition domain.

Potential applications include:

- Consumer nutrition awareness.
 - Healthy food recommendation systems.
 - Nutritional benchmarking.
 - Decision support for diet planning.
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Tools & Technologies

- Python
 - Pandas
 - NumPy
 - MySQL
 - SQLAlchemy
 - Power BI
 - Git & GitHub
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Conclusion

The Food Nutrition & Health Risk Analytics System successfully analyzed 1,028 food items to evaluate their nutritional quality and potential health risks. By developing custom metrics such as Nutrition Density and Health Risk Score, the project provided a more comprehensive assessment of food healthiness than individual nutritional values alone.

The analysis identified foods such as Maa Chaane Ki Dal, Cracked Wheat and Green Gram Dal Premix, and Dhokla as some of the healthiest options in the dataset, while products like Butter Icing and Glace Icing were found to have the highest health risk scores. Additionally, the majority of foods were classified as low risk, indicating a generally favorable nutritional profile within the dataset.

Overall, this project demonstrates how data analytics, SQL, and Power BI can be used to transform raw nutritional data into meaningful insights that support healthier food choices and nutrition-focused decision-making.

Future Enhancements

- Include additional nutritional factors such as sodium and saturated fats in the risk assessment model.
- Develop machine learning models for automated health risk prediction.
- Build a personalized food recommendation system based on nutritional goals.
- Integrate real-time nutrition data from external APIs.
- Deploy the dashboard as a web application for broader accessibility.